NET Centric Computing

Course Title: NET Centric Computing
Course no: BIT351
Nature of the Course: Theory + Lab
Semester: VI

Full Marks: 60+20+20
Pass Marks: 24 +8+8
Credit Hrs: 3

Course Description:
The course covers the concepts of cross-platform web application development using the ASP.NET Core MVC framework using C# programming Language.

Course Objectives:
The objective of this course is to understand the theoretical foundation as well as its practical aspects of ASP.NET Core web application framework and C# language features.

Course Contents:

Unit 1: Language Preliminaries (8Hrs.)
Overview of .Net framework, Compilation and execution of .Net applications, Constructor, Properties, Arrays and String, Indexers, Inheritance, use of "base" keyword, Method hiding and overriding, applying polymorphism in code extensibility, abstract class sealed class, interface, Delegate and Events, Partial class, Collections, Generics, File IO, Try statements and Exceptions, Attributes: Attribute Classes, Named and Positional Attribute Parameters, Attribute Targets, Specifying Multiple Attributes

Unit 2: Introduction to ASP.NET (3 Hrs.)
.NET andASP.NET frameworks: .NET, .NET Core, Mono, ASP.NET Web Forms, ASP.NET MVC, ASP.NET Web API, ASP.NET Core, .NET Architecture and Design Principles, Compilation and Execution of .NET applications: CLI, MSIL and CLR, .NET Core in detail, .NET CLI: build, run, test and deploy .NET Core Applications

Unit 3: HTTP and ASP.NET Core (3 Hrs.)
HTTP, Request and Response Message Format, Common web application architectures, MVC Pattern, ASP.NET Core Architecture Overview, Projects, and Conventions, ASP.NET and ASP.NET MVC

Unit 4: Creating ASP.NET core MVC applications (7 Hrs.)
Setting up the Environment, Controllers and Actions: Create Controllers, Create Actions and Action Results Types, Rendering HTML with Views: Razor Syntax, Understanding Tag Helpers, Models: Binding and Validations, URL Routing and features, Web API Applications: API Controllers, JSON, Dependency Injection and IOC containers

Unit 5: Working with Database (6 Hrs.)
ADO.NET basics: Connection, Command, Reader and Adapter classes, Entity Framework (EF) Core, Object-Relational Mapper (ORM), Adding EF Core to an application: Choosing database provider, data models and data context, Querying and Saving data to database: Create, read, update and delete records

Unit 6: State Management on ASP.NET Core Application (4 Hrs.)
State Management on stateless HTTP, Server-side strategies: Session State, TempData, Using HttpContext, Cache Client-side strategies: Cookies, Query Strings, Hidden Fields

Unit 7: Client-side Development in ASP.NET Core (4 Hrs.)
Common client-side web technologies, JQuery, Forms and Validation, Single Page Application (SPA) Frameworks: Angular, React

Unit 8: Building Applications with React and ASP.NET Core (3 Hrs.)
Creating Forms in React, Making REST API Calls from React to ASP.NET Core for integration

Unit 9: Securing in ASP.NET Core Application (5 Hrs.)
Authentication: ASP.NET Core Identity, Adding authentication to apps and identity service configurations, Authorization: Roles, Claims and Policies, Securing Controllers and Action Methods, Common Vulnerabilities: Cross-site Scripting attacks, SQL Injection attacks, Cross-site Request Forgery (CSRF), Open Redirect Attacks

Unit 10: Hosting and Deploying ASP.NET Core Application (2 Hrs.)
App Servers and Hosting models: IIS, Nginx, Apache, ASP.NET Core Module, Kestrel, Docker and Containerization, Publish toAzure cloud

References:

1. C# 8.0 and .NET Core 3.0 – Modern Cross-Platform Development, Fourth Edition, by Mark J. Price, 2019
2. ASP.NET Core in Action, by Andrew Lock, 2018
3. Learning ASP.NET Core 2.0, Michel Bruchet, Jason De Oliveira, 2017
4. Learn ASP.NET Core 3 - Second Edition, Kenneth Yamikani Fukizi, Jason De Oliveira, Michel Bruchet, 2019
5. ASP.NET Core 5 and React By Carl Rippon

Laboratory works:
Students will have to complete a project covering most of the features of above course using ASP.NET Core and C#.

Homework Assignments:
Homework assignments can be given according to the course covered throughout the semester.

Computer Usage:
Windows PC or workstation installed with Latest Visual Studio Code or Visual Studio, .NET core SDK (3.0 or above) and Database: SQL Server.

Database Administration

Course Title: Database Administration
Course No: BIT352
Nature of the Course: Theory + Lab
Semester: VI

Full Marks: 60+ 20+20
Pass Marks: 24+8+8
Credit Hrs: 3

Course Description:

This course introduces the database administration techniques in Oracle. Most of the DBA's tasks are covered. Topics covered include principles of DBA Roles, Oracle Database Architecture and storage management, Database backup, restoration and recovery, connectivity and user management for database security, Tuning of database and overall DB administration which could be useful for administrator in the future.

Course Objectives:

To understand the basic role, task and responsibilities of Database Administrator, understand the Oracle database architecture, be able to install and configure an Oracle Database, be able to create database objects like tables, views, indexes etc. and able to write PL/SQL Procedures, be able to administer the Oracle Database, create and manage storage structures, create and manage the users, be able to perform backup and recovery, tuning the oracle database for the better performance.

Course Contents:

Unit 1: Introduction Administration Basics (7Hrs.)

1.1. Introduction: Overview of the Oracle Database, Relational Database Concepts, Overview of Oracle Database Architecture, Overview of Oracle Multitenant Architecture, Overview of Oracle Data Guard, Oracle RAC, Oracle ASM, Oracle Cloud, Database Administration Tasks.
1.2. Oracle Architecture: Oracle Database Architecture, the Oracle Database Instance, Memory Structures, Process Architecture and Structures, Server and Client Processes, Database Storage Architecture, Connect to the Oracle Database Instance.
1.3. Administer Database Instance: Administrative Tools Available to a DBA, Use SQL*Plus in Oracle to Manage a Database Instance, use SQL Developer to Manage a Database Instance, Administer the Database using Enterprise Manager (EM), Overview of the Enterprise Manager Framework, Enterprise Manager Cloud Control, Initialization Parameter Files

Unit 2: Network Environment and Storage Structures (5 Hrs.)

2.1. Configuring Network Environment: Overview of Network Configuration, Oracle Net Listener Configuration and Management, Oracle Net Naming Methods, Tools for Configuring and Managing the Oracle Network, using the Net Configuration Assistant,

Configure Client Connections with Net Manager, View Listener Configuration, Start and Stop the Oracle Listener, use TNSPING to Test Oracle Net Connectivity, Connect to the Database, Configure Net Services with Enterprise Manager.

2.2. Storage Structures: Overview of Data Storage, the Database Block, Overview of Tablespaces and Datafiles, use Enterprise Manager to view the Storage Structure of the Database, Create and Alter Tablespace Commands, Temporary Tablespaces, Create and Manage Datafiles, use OMF, Drop Tablespaces and Datafiles

Unit 3: Users and Privileges (5 Hrs.)

3.1. Manage Users: Predefined Database Administration Accounts, User Accounts, Create a User Account, User Authentication, Change a User's Password, Manage a User Account, Drop a User Account, Monitor User Information, Terminate User Sessions, Create a Schema Only Account

3.2. Mange Privileges: Database Access, Oracle Supplied Roles, System and Object Level Privileges, the Grant and Revoke Commands, Create, Modify and Drop Roles, Use Predefined Roles

Unit 4: Profiles, Resources, and Auditing (4 Hrs.)

4.1. Profiles And Resources: Overview of User Profiles, Profile Resource Parameters, Create Profile Command, Manage Passwords with Profiles, Control Resource Usage with Profiles, Maintain Profiles

4.2. Database Auditing: Overview of Database Security, Overview of Database Auditing, Security Compliance, Standard Auditing, Unified Audit Trail, Separation of Audit Responsibilities with The AUDIT_ADMIN And AUDIT_VIEWER Roles, Configure the Audit Trail, Specify Audit Options.

Unit 5: Concurrency, Backup and Recovery (7 Hrs.)

5.1. Concurrency: Levels of Locking in Oracle, Methods Used to Acquire Locks, Data Concurrency, Possible Causes of Contention, DML Locks, Prevent Locking Problems, Detect Lock Contention, Resolve Conflicts

5.2. Undo Management: Undo Data Overview, Monitor and Administer Undo, Configure Undo Retention, Switch Undo Tablespaces, Specify the Retention Period, Guarantee Undo Retention, Retention Period tor Flashback Queries, View Undo Space Information, use the Undo Advisor, Size the Undo Tablespace, Alter an Undo Tablespace to a Fixed Size

5.3. Backup And Recovery: Oracle Backup Solutions, Oracle Suggested Backup Strategy, Overview of Database Backup, Restore and Recover, Flashback Technology, Types of Failure, Instance Recovery, Tune Instance Recovery, The MTTR Advisor, Media Failure, Configure a Database for Recoverability

Unit 6: Database Maintenance and Performance Management (4 Hrs.)

6.1. Database Maintenance: Overview, View the Alert Log, the Automatic Workload Repository, Statistic Levels, the Automatic Database Diagnostic Monitoring, Monitor an Oracle Database, use the Advisors, Set Up Notification Rules
6.2. Performance Management: Tuning Information Sources, Performance Monitoring, Tuning Activities, Performance Planning, Instance Tuning, Performance Tuning Methodology, Performance Tuning Data, Monitoring Performance, Managing Memory, Manage Private Temporary Tables

Unit 7: Tuning and Moving (5 Hrs.)

7.1. Tuning: SQL Tuning, Cancel a SQL Statement in a Session, the Oracle Optimizer, SQL Plan Directives, Adaptive Execution Plans, SQL Advisors, Automatic SQL Tuning Results, Implement Automatic Tuning Recommendations, SQL Tuning Advisor
7.2. Moving Data: Create Directory Objects, Data Pump Architecture, Data Pump Data Dictionary Views, Data Pump Interactive Mode, Data Pump API, use Data Pump to Export Data, Use Data Pump to Import Data, Overview of SQL Loader, Command Line Parameters, Record Filtering, Control File Keywords, Data files, SQL Loader Data Paths, External Tables

Unit 8: Managing Resources and Scheduling Tasks (4 Hrs.)

8.1 Managing Resources: Overview of the Database Resource Manager, use the Resource Manager, Create Resource Plans, the Default Maintenance Resource Manager Plan, Create Resource Plan Directives, Allocate Resources for Resource Plans
8.2 Scheduling Tasks: Introduction to The Scheduler, Access Rights, Scheduler Components and Workflow, Create a Job, Job Classes, use Time Based, Event Based Schedules, Create an Event Based Schedule

Unit 9: Space Management and Oracle Support (4 Hrs.)

9.1. Managing Space: Overview of Space Management, Block Space Management within Segments, Segment Types, Allocate Extents, Allocate Space, Row Chaining and Migration, Create Tables without Segments
9.2. Oracle Support: The Enterprise Manager Support Workbench, Register for Security Updates, Work with Oracle Support, My Oracle Support Integration, Log Service Requests, Manage Patches, Apply a Patch

Laboratory Works:

Student should prepare lab sheet for most of the units in the syllabus. They should practice, design database and implement database administration activities that demonstrates different concepts discussed is the class.

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References:

1. Oracle Database 19c DBA By Examples: Installation and Administration, Ravindra Gupta, 2021.
2. https://docs.oracle.com/en/database/oracle/oracle-database/19/administration.html
3. Pro Oracle Database 18c Administration: Manage and Safeguard Your Organization's Data, Michelle Malcher and Darl Kuhn, Third Edition.
4. Oracle Database 12c DBA Handbook, Manage a Scalable, Secure Oracle Enterprise Database Environment, Bob Bryla.
5. Oracle DBA Mentor: Succeeding as an Oracle Database Administrator, Brian Peasland.

Management Information System

Course Title: Management Information System
Course No: BIT353
Nature of the Course: Theory + Lab
Semester: VI

Full Marks: 60 + 20 + 20
Pass Marks: 24 + 8 + 8
Credit Hrs: 3

Course Description:
This course covers different concepts of management information system including information system in the global business, collaboration and tools, different types of information systems, ethical and social issues, concept of business intelligence, enterprise applications, and decision making.

Course Objectives:
The main objective of this course is to provide different concepts of management information system to manage business organizations and efficient decision making.

Course Contents:

Unit 1: Information Systems in Global Business Today (3 Hrs.)
What's New in Management Information System?, Globalization Challenges and Opportunities: A Flattened World, The Emerging Digital Firm, Strategic Business Objectives of Information System, Information System and its Dimensions, Business Perspective on Information Systems, Complementary Assets, Academic Disciplines used to Study Information Systems, Sociotechnical Systems

Unit 2: Global Business and Collaboration (6 Hrs.)
Business Process and their Relationship to Information System, Systems for Different Management Groups, Systems for Linking the Enterprise, E-business, E-commerce, and E-government, Collaboration, Social Business, and Benefits, Building Collaborative Culture and Business Processes, Tools and Technologies for Collaboration and Social Business, Information Systems Function in a Business – Information Systems Department, Organizing Information Systems Function

Unit 3: Information Systems, Organization, and Strategy (8 Hrs.)
Organization and its Features, Impact of Information System on Organizations – Economic Impacts, Organization and Behavioral Impacts, Internet and Organizations, and Implications for Design and Understanding of Information Systems, Porter's Competitive Forces Model, Dealing with Competitive Forces using Information Systems, Internet's Impact on Competitive Advantage and Business Value Chain Model, Challenges Posed by Strategic Information System

Unit 4: Ethical and Social Issues in Information Systems (7 Hrs.)
Ethical, Social, and Political Issues raised by Information Systems – Model, Five Moral Dimensions, Key Technology Trends that Raise Ethical Issues, Responsibility, Accountability, and Liability, Ethical Analysis, Candidate Ethical Principles, Professional Codes of Conduct,

Real World Ethical Dilemmas, Information Right, Property Right, Accountability, Liability, System Quality, Quality of Life, Health Risks

Unit 5: Foundations of Business Intelligence (8 Hrs.)
File Organization and Problems with Traditional File Environment, DBMS and its Capabilities, Designing Databases, Nonrelational Databases, Cloud Databases, and Blockchain, Tools and Technologies for accessing Information from Databases to Improve Business Performance and Decision Making, Big Data Challenges, Business Intelligence Infrastructure, Analytical Tools, Assuring Data Quality

Unit 6: Key System Applications for Digital Age (7 Hrs.)
Enterprise Systems, Enterprise Software and Business Value of Enterprise Systems, Supply Chain Management System and Software, Business Value of Supply Chain Management System, Customer Relationship Management System and Software, Business Value of CRM, Challenges of Enterprise Applications, Role of Knowledge Management System in Business, Artificial Intelligence and Machine Learning, Uses of AI in Business, Knowledge Work Systems

Unit 7: Enhancing Decision Making (6 Hrs.)
Types of Decisions, Decision Making Process, Business Value of Improved Decision Making, Information System, Management Activities, and Management Decision Making, Business Intelligence and Business Analytics Support for Decision Making, Decision Support for Operational, Middle and Senior Management with Business Intelligence

References:

1. Management Information Systems: Managing the Digital Firm, Kenneth C. Laudon and Jane P. Laudon, 17th Edition, Pearson, 2022.
2. Information Technology Essentials: Introduction to Information Systems Volume 1, Eric Frick, 2017.
3. Management Information Systems, James O'Brien and George Marakas, 10th Edition, McGraw Hill.

Research Methodology

Course Title: Research Methodology **Full Marks:** 60+40
Course No: RSM354 **Pass Marks:** 24+16
Nature of the Course: Theory **Credit Hrs:** 3
Semester: VI

Course Description:
This course introduces the concepts of research methodology. It also emphasizes on formulating research questions, conducting literature review, data collection, data analysis based on research questions and other various ethical issues, plagiarism.

Course Objectives:
This course focuses on the concept of research methodology; research design, formulation of research questions, process of review of literature, methods of data collection, measurement issues, ethical issues, plagiarism, data analysis and interpretations. After completion of the course, students will also be able to prepare a mini research report following standard notions of research methodology.

Course Contents:

Unit 1: Introduction to Research Methodology (4 Hrs.)
Meaning, objectives, motivation of research, concept of deductive and inductive theory, characteristics of scientific method, understanding of concept, construct, variable; research process

Unit 2: Research Design (6 Hrs.)
Concept of research design and its importance, features of a good research design, exploratory research design, descriptive research design, experimental research design, quantitative & qualitative research, empirical research, case study, applications of different research designs and their limitations, concept of dependent variable, independent variable and assessment of their relationships, issue of generalization of research findings

Unit 3: Identification and Formulation of Research Question (5 Hrs.)
Identification of research questions and its formulation, research objectives, statement of the problem, setting up the research hypothesis, sources of hypothesis and its utilities

Unit 4: Review of Literature (10 Hrs.)
Concept of review of literature, process of reviewing books, journal articles, reports; citation and referencing, an overview of different formats of citation and referencing, citation of books, reports, journal articles in American Psychological Association(APA) format, preparation of a small report based on review work on some relevant specific topic,use of encyclopedias, research guides, handbook, academic databases for computer Science and information technology field, use of reference management software such as Zotero/Mendeley, paper formatting such as LaTeX/MS Office, iThenticate Software for detection of Plagiarism

Unit 5: Methods of Data Collection and Sampling Techniques (7 Hrs.)

Primary and secondary data, method of collecting primary data, preparation of questionnaire, type of questions, characteristics of good questionnaire, concept of sampling, population, sampling frame, sampling and non-sampling error, probability and non-probability sampling, brief overview of simple random sampling, stratified sampling, cluster sampling, systematic sampling, multistage sampling and their practical applications in research problems, sample size estimation for estimating mean and proportion

Unit 6: Measurement and Scaling (3 Hrs.)

Concept of measurement, different levels of measurement, measurement of variables in likert scales, the issue of validity and reliability in research

Unit 7: Data Analysis and Report preparation (10 Hrs.)

Presentation of data through frequency tables, bar charts, pie charts, percentages, cross tabulations, summary measures, use of inferential statistical analysis in bivariate analysis(t-test, chi-square test, correlations, etc.) and concept of multivariate analysis, format of academic research report, preparation of a mini-research report following standard research methodology; ethical issues related to use of data, plagiarism, informed consent, self-privacy while collecting primary data, etc.

References:

1. Bryman A. & Bell E.(2015). *Business Research Methods*. Oxford University Press
2. Day R. A. &Gastel B. (2016).*How to write and publish a scientific paper*. (8thed.). Greenwood Publication.
3. Kerlinger F.N. (2004). *Foundations of Behavioral Research*.Surgeet Publications, New Delhi, India
4. Kothari C.R. (2004). *Research Methodology*. New Age International (P) Limited, Publishers, India
5. Kumar, Ranjit (2011). *Research Methodology*. SAGE Publications Pvt. Limited, India.

Geographical Information System

Course Title: Geographical Information System
Course Code: BIT355
Nature of the course: Theory + Lab
Semester: VI

Full Marks: 60+20+20
Pass Marks: 24+8+8
Credit Hrs: 3

Course Description:
The course covers about basic concepts of capturing and mapping real world, different spatial data model and structure, modeling and database design, spatial data manipulation, analysis and visualization, overview of open GIS, open source GIS data and application of GIS in different fields for communication and decision making process.

Course Objectives:
The main objective of this course is to provide students with knowledge of fundamental concepts of geographic information system.

Unit 1: The basics of geographic information system (5 Hrs.)
Definition of GIS, GIS Components and Functions, GIS and Cartography, Evolution of GIS in Relation to Computational and Information Technology, Benefits of GIS, GIS Application Areas, Introduction to Different GIS Software

Unit 2: Mapping the Real World with Vector and Raster Data (8 Hrs.)
Modeling and Representing the Real World, Vector Data Model, Vector Data Types, Vector Topology and Data Formats, Building Basic Topology, Comparison of Shapefile, Coverage and Geodatabase, Geo-relational and Object-relational Vector Data Models, TIN Data Model, Raster Data Model, Basic Elements of Raster Data, Raster Data Formats, Georeferencing Imagery, Creating Vector Data from Digitization, Map Scale, Precision and Accuracy, Spatial Resolution and Data Volume, Vector vs. Raster Data Model, Map Design and Layout, Map Elements

Unit 3: Data Acquisition Techniques in GIS (8 Hrs.)
Spatial and non-spatial data, Primary and Secondary Data Capture Techniques, Concepts of Field Survey for Large Scale Mapping, Scanning and Digitizing, Introduction to global navigation Satellite System (GNSS), Remote Sensing Technology, Satellite Orbits, Aerial Photography, Photogrammetry, Other emerging imaging techniques, Conversion and Integration of GPS and RS data with GIS

Unit 4: Map Projection and Coordinate System (6 Hrs.)
Concepts of Longitude and Latitude, Geographic and Projected Coordinate Systems, Mathematical Model of Earth, Geoid and Ellipsoids, Horizontal and Vertical Datum, Changing Projection and Coordinate Systems, Types of Projection Systems, Projection Parameters, Working with Map Projection

Unit 5: Vector Data Analysis (7 Hrs.)

Filtering Data using Queries, Vector Overlay Analysis, Geoprocessing Functions, Proximity and Buffer Analysis, Route Optimization and Shortest Path, Mapping Non-spatial Data, Creating ModelBuilder

Unit 6: Raster Data Analysis (7 Hrs.)

Overview of Raster Overlay, Map Algebra, Raster Calculator, Overlaying Raster, Introduction to Interpolation and Spatial Autocorrelation, DEM, Spatial Statistics, Local Raster Operations, Focal Raster Operations, Zonal Raster Operations, Global Raster Operations, Raster Weighted Overlay, Mosaic and Aggregate tools, Distance Measurement

Unit 7: Open GIS and Applications (4 Hrs.)

Introduction of open concept in GIS, Open source software for spatial data analysis, Open Vs Commercial GIS Program, GIS programming and customization, Python script tools, Customizing QGIS with Python, Web Based GIS system, Open source GIS data, Introduction to Google Earth Engine and Applications, GIS application case studies

Laboratory works: The lab should cover at least the concepts given in the chapters

References:

1. P. Lo and Albert K.W. Yeung, Concepts and Techniques of Geographic Information Systems, Pearson Prentice Hall
2. K. T. Chang, Introduction to geographic information systems. Ninth edition, Boston: McGraw-Hill.
3. Longley, P.A., Goodchild, M.F., Maguire, D.J. and Rhind, D.W., Geographic information systems and science. John Wiley & Sons.
4. Huisman, Otto, and Rolf A. de By. "Principles of geographic information systems." ITC Educational Textbook Series 1 (2009): 17.
5. Kerle, Norman, Lucas LF Janssen, and Gerrit C. Huurneman. Principles of remote sensing." ITC, Educational textbook series 2 (2004): 250.
6. Burrough, P. A., McDonnell, R. A., & Lloyd, C. D. (2015). Principles of geographical information systems. Oxford university press.
7. ESRI guide to GIS analysis Andy Mitchell, ESRI press, Red lands

Multimedia Computing

Course Title: Multimedia Computing
Course No: BIT356
Nature of the Course: Theory + Lab
Semester: VI

Full Marks: 60 + 20 + 20
Pass Marks: 24 + 8 + 8
Credit Hrs: 3

Course Description: This course familiarizes students with the concepts of multimedia computing including sound, image, video, animations, data compression, and multimedia applications.

Course Objectives: The main objective of this course is to provide knowledge of different concepts of multimedia computing and their applications.

Course Contents:

Unit 1: Introduction (2 Hrs.)
Multimedia Systems; Components of multimedia; Multimedia System, its characteristics and properties; Structure of Multimedia; Applications

Unit 2: Text / Sound / Audio System (7 Hrs.)
Text: Types of Text, Ways to Present Text, Aspects of Text Design, Character, Character Set, Codes, Unicode, Encryption
Sound / Audio: Basic Sound Concepts, Types of Sound, Digitizing Sound, Audio Formats, Audio tools, MIDI; Speech Generation, Speech Analysis and Speech Transmission

Unit 3: Image and Video (6 Hrs.)
Image: Digital Image Representation, Uses of Images, Formats, Image Color Scheme, Image Enhancement; Image Synthesis, Analysis and Transmission
Video: Analogue and Digital Video, Recording Formats and Standards (JPEG, MPEG, H.261) Transmission of Video Signals, Video Capture

Unit 4: Video and Animation (8 Hrs.)
Digital Video; Video Signal Representation; Computer Video Format; Computer-Based Animation; Animation Language; Timeline and Frame based animation; Timeline and Tween-Based Animation; Methods of controlling Animation; Display of Animation; Transmission of Animation

Unit 5: Data Compression (5 Hrs.)
Introduction; basics of compression; storage space; need of coding; Lossy and Lossless Compression Techniques; Source, Entropy and Hybrid Coding; Lossy Sequential DCT – based mode; Expanded Lossy DCT-based mode; JPEG and MPEG Compression

Unit 6: Abstraction for Programming (5 Hrs.)
Introduction to abstraction levels; Libraries; System Software; Toolkits; Higher Programming Languages; Object Oriented Approaches

Unit 7: Multimedia Design (4 Hrs.)
Development Phases and Development Teams; Analysis, Design, Development and Implementation Phase; Evaluation and Testing Phase

Unit 8: User Interfaces (4 Hrs.)
Basic Design Issues; Video and Audio at the User Interface; User-Friendliness as the Primary Goal

Unit 9: Multimedia Application (4 Hrs.)
Media Preparation and Composition; Media Integration and Communication; Media Entertainment; Telemedicine; E-learning; Digital Video Editing and Production Systems; Video Conferencing; Video-on-demand

Laboratory Works:
The laboratory works should focus on writing programs of different concepts of multimedia computing

References:

1. Multimedia: Computing, Communications and Applications, Ralf Steinmetz and Klara Nahrstedt, Pearson Education Asia
2. Multimedia Communications, Applications, Networks, Protocols and Standards, Fred Halsall, Pearson Education Asia
3. Multimedia Systems, John F. Koegel Buford, Pearson Education Asia
4. Multimedia Fundamentals: Vol. 1- Media Coding and Content Processing, Ralf Steinmetz and Klara Nahrstedt, PHI.

Wireless Networking

Course Title: Wireless Networking
Course No: BIT357
Nature of the Course: Theory + Lab
Semester: VI

Full Marks: 60+ 20+20
Pass Marks: 24+8+8
Credit Hrs: 3

Course Description:

The course addresses the fundamentals of wireless communications and provides an overview of existing and emerging wireless communications networks. It covers radio propagation and fading models, fundamentals of cellular communications, multiple access technologies, and various wireless networks, including past and future generation networks. Simulation of wireless systems under different channel environments will be integral part of this course.

Course Objectives:

To characterize fading multi-path radio channels, describe different types of diversity for mobile radio channels, explain propagation models for mobile and portable wireless communication, analyze simple wireless networks in terms of coverage and capacity, discuss multiple access techniques and slandered, describe mobility management strategies and traffic calculation, describe concept of mobile IP, protocols and routing in ad-hoc network.

Course Contents:

Unit 1: Overview of wireless communications and systems (2 Hrs.)

1.1. Introduction to Wireless Communications, Challenges in Wireless Communication Networks, Cellular Systems from 1G to 3G , Wireless 4G and 5G Systems

Unit 2: Wireless Channel Characterization (7 Hrs.)

2.1. Multipath Propagation Environment, Small Scale Fading, Fading Effects due to Multipath Time Delay Spread, Fading Effects due to Doppler Spread
2.2. Channel Models, Fading Models: Rayleigh Fading Distribution, Ricean Fading Distribution
2.3. Large Scale Path-Loss and Shadowing, Free-Space Path Loss Model, Propagation Over Reflecting Surface (Smoothing Plane), Long Distance Path loss with Shadowing: Okumura-Hara Path Loss Model

Unit 3: Band Pass Transmission Technique for Mobile Radio (9 Hrs.)

3.1. An overview of Digital Communication, Pulse Shaping Technique, Nyquist Pulse Shaping, Raised Cosine Roll-off Filter
3.2. Modulation Techniques for Mobile Radio, Analog and Digital Modulation – An overview, Criteria of Choosing Modulation Schemes, Geometric Representation of Modulated signal, Power Spectral Density, Probability of Error
3.3. Digital Modulation Techniques, Digital Linear Modulation (BPSK, DPSK, QPSK)
3.4. Minimum Shift Keying (MSK), Gaussian Minimum Shift Keying (GMSK), M-array (MPSK, MFSK, QAM and OFDM) Modulation and Demodulation

Unit 4: Equalization, Diversity and Channel Coding (4 Hrs.)

4.1. Basics of Equalization, Equalization in Communications Receivers, Linear Equalizers, Non-Linear Equalization, Decision Feedback and Maximum Likelihood Sequence Estimation Equalizations

4.2. Adaptive Equalization Algorithms, Zero Forcing, Least Mean Square, Recursive Least Squares Algorithms, Fractionally Spaced Equalizers

4.3. Diversity Methods, Advantages of Diversity, Basic Definitions, Space Diversity, Reception Methods (Selection, Feedback, Maximum Ratio and Equal Gain Diversity), Polarization, Frequency and Time Diversity, RAKE Receivers and Interleaving

Unit 5: Fundamental of Cellular Network (6 Hrs.)

5.1. Transmission Control Protocol (TCP), User Datagram Protocol (UDP), Ports, IP Address Network Classes in JDK

5.2. Socket Programming using TCP, Socket Programming using UDP, working with URL's, working with URL Connection Class

Unit 6: Multiple Access in Wireless Network (6 Hrs.)

6.1. Frequency Division Multiple Access (FDMA) Principle and Application, Time Division Multiple Access (TDMA) Principles and Applications

6.2. Spread Spectrum Multiple Access, Frequency Hopped Multiple Access, Code Division Multiple Access, hybrid spread spectrum multiple access techniques, Space Division Multiple Access, Standards for Wireless Local Area Networks

Unit 7: Mobility Management in Wireless Network (5 Hrs.)

7.1. Introduction to Mobility Management, Call Admission Control (CAC), Handoff Management, Handoff Strategies, Handoff Types

7.2. Location Management for Cellular Network, Location Management for PCS Network, Traffic Calculation

Unit 8: Wireless Internetworking (6 Hrs.)

8.1. Introduction to Internetworking for Wireless Networks

8.2. Concept of mobile IP, Architecture and Operation, Tunneling in mobile IP.

8.3. Mobility in IPv6, Transmission Control Protocol (TCP), Wireless Application Protocol (WAP), Wireless Markup Language (WML), Mobile AD HOC Network (MANET), ADHOC Routing Protocols

Laboratory Works:

Students should write programs and prepare lab sheet for most of the units in the syllabus. Majorly, students should practice design and implementation of wireless network. Students are advised to implement the modulator de-modulator, frequency planning, channel assignment as well as routing algorithms used in wireless network. Students are advised to use simulators. Students are advised to visit the mobile service operators, network service providers, internet

service providers and prepare the report including architecture, service, and functioning of the wireless network.

References:

1. Jon W. Mark and Weihua Zhuang, Wireless Communication and Networking, Prentice Hall
2. K. Feher, Wireless Digital Communications, Prentice Hall
3. T. Rappaport, Wireless Communications, Prentice Hall
4. J. Schiller, Mobile Communications, Pearson